Course notes

Primitives over an interval

It is admitted at this stage that any function continues over an interval admits antiderivatives; the connection with Chapter 8 will be prepared for this purpose. f over an interval I is a detritable function F such that $F' = f$ on I. If F is a antiderivative, all antiderivatives on I are $F + C, C \in \mathbb{R}$.

Demonstration. If $F' = G' = f$, then $(F - G)' = 0$ on the interval; $F - G$ is therefore constant. Conversely, $(F + C)' = f$. The interval hypothesis is essential: on a disjunct domain, the constants may differ according to the intervals.

Table of antiderivatives · inverse reading of derivatives

$f(x)$	$F(x)$
$x^n \ (n \in \mathbb{Z},\ n \neq -1)$	$\frac{x^{n+1}}{n+1}$
$\frac{1}{x}$	$\ln \lvert x \rvert$ over an interval avoiding 0
$\frac{1}{\sqrt{x}}$	$2\sqrt{x} \quad (x > 0)$
e^x	e^x
$\cos x$	$\sin x$
$\sin x$	$-\cos x$

The additive constant can always be added. The last two lines will be reinvested After the study of trigonometry.

Recognition of membership

For a function u Drifting, antiderivatives of $u'u^n$, $u'e^u$, $\frac{u'}{u}$ are respectively $\frac{u^{n+1}}{n+1}\ (n \neq -1)$, e^u, $\ln \lvert u \rvert$, on intervals where expressions have a meaning. Linearity allows to process sums.

Example. A antiderivative of 6 $x(x^2+1)^2$ is $(x^2+1)^3$; a antiderivative of e^{3x} is $\frac{e^{3x}}{3}$. To impose $F(0) = 2$ in the latter case, take $F(x) = \frac{e^{3x}}{3} + \frac{5}{3}$.

Differential equations

A differential equation connects an unknown function to its derivatives. over a specified interval.

Common solutions

$y' = ay$ a for solutions $y(x) = Ce^{ax}$. For $a \neq 0$, $y' = ay + b$ a for solutions

$$y(x) = Ce^{ax} - \frac{b}{a}.$$

If $a = 0$, the equation becomes $y' = b$ and $y(x) = bx + C$. If g is a particular known solution of $y' = ay + f(x)$, solutions are $y = g + Ce^{ax}$.

Demonstration for $y' = ay$. If y is a solution, $z(x) = e^{-ax}y(x)$ checks $z' = e^{-ax}(y' - ay) = 0$; $z = C$. Conversely, the derivative of Ce^{ax} is aCe^{ax}. For $y' = ay + b$, subtract the constant solution $-\frac{b}{a}$ the same equation. For a second variable member, do the same with g.

Example. $y' = -2y + 6$ and $y(0) = 1$ give $y = 3 + Ce^{-2x}$then $C = -2$. Solution 3 $-2e^{-2x}$ growth to balance 3 when $x \to +\infty$.

Method of production

For $y' = ay + b$, with one step $h > 0$, we're approaching $y(t_k)$ by Y_k by $t_{k+1} = t_k + h$, $Y_{k+1} = Y_k + h(aY_k + b)$. It is an approximation using tangent, not equality on the exact solution. A smaller step can improve approximation; numerically control error on a solution known.

Dive backwards and model an evolution

7.1 Recognize and control a antiderivative

First use: S52.

Definition · Primitive over an interval

A F function is a antiderivative f on I if F is driftable on I and $F' = f$ I. Interval is part of the data. Finding a antiderivative is an inverse calculation; the derivation allows to Check it out.

Example Solved · Correct a Factor

For $f(x) = 3e^{2x}$, $F(x) = \frac{3}{2}e^{2x}$ is appropriate because $F'(x) = \frac{3}{2} \times 2e^{2x} = 3e^{2x}$. For $g(x) = 2x/(1+x^2)$ on $\mathbb{R}$, the antiderivative $G(x) = \ln(1+x^2)$ uses a positive argument. All antiderivatives obtain by adding a constant.

A1 · Immediate application Determine a antiderivative of 4 $x^3 - 2x + 1$, of e^{-3x} and 1 $/(2x+1)$ on a suitable interval. Check by bypass.

7.2. A constant, a vertical translation

First use: S52.

Demonstration. If $F' = G' = f$ over interval I, then $(F-G)' = 0$ on I. A derivative function zero over an interval is constant; therefore $F - G = C$. On a domain consisting of several intervals Disjointed, the constant can be different over each interval. □

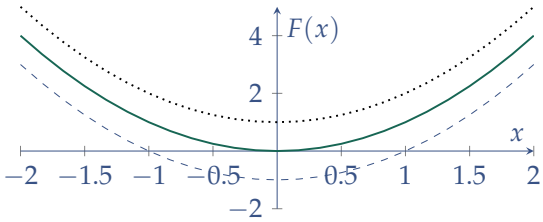


Figure 7.1. Three antiderivatives of $2x$: On the same apses, the tangents have the same slopes.

Solved Example · A value sets the translation

The antiderivatives of $2x$ are $x^2 + C$. Imposing F(0)=1 selects $F(x) = x^2 - 1$. The graph is Translating a unit downwards without changing the slopes.

A2 · Partial guidance Determine the antiderivative of $2x$ which is worth 3 in 1. Then explain why all the antiderivatives of $1/x$ on $\mathbb{R} \setminus \{0\}$ are not described by a single constant common to both Half straight.

7.3. Unknown is a function

First use: S55.

Ownership · Equation $y' = ay$

For $a \in \mathbb{R}$, solutions on $\mathbb{R}$ are $y(t) = Ce^{at}$, $C \in \mathbb{R}$. An initial value $y(0) = y_0$ imposes $C = y_0$.

Demonstration. A function of this form is well solution. Conversely, if it is solution, $z(t) = y(t)e^{-at}$ a for derivative $(y' - ay)e^{-at} = 0$, so is constant over the interval. $y(t) = Ce^{at}$. The evidence checks the solutions and shows that none is missing. □

Example resolved · A condition in another instant

For $y' = 3y$ and y(1)=2, $y(t) = Ce^{3t}$then $Ce^3 = 2$. Therefore $C = 2e^{-3}$ and $y(t) = 2e^{3(t-1)}$. The constant is equal to the value imposed only if it is given in 0.

A3 · Immediate application Solve $y' = -2y$ with $y(0) = 5$. Is the number 0 a solution possible differential equation alone ? Of this problem with initial condition ?

7.4 Subtract a known solution

First use: S55 and S58.

Example resolved · Second variable member

For $y' = -y + 2t + 1$, seek a particular solution $p(t) = at + b$ led to $a = -at - b + 2t + 1$. Identify the coefficients given $a = 2$, $b = -1$. If y is a solution, $z = y - p$ checks $z' = -z$. With $y(0) = 3$, we get $z(0) = 4$, so $y(t) = 2t - 1 + 4e^{-t}$.

A4 · Partial guidance Solve $y' = 2y - 4$ with $y(0) = 3$ by subtracting a constant solution. Check the formula and initial value.

A5 · Repair Evidence One student checked that a function satisfies a differential equation, and then "I have found all the solutions." Why is this conclusion insufficient? Illustrate with $y' = y$.

7.5 Discrete, Continuous and Approximative Comparison

First use: S55 and S58.

Example Solved · Cooling

The model $T' = -0{,}1(T - 20)$, $T(0) = 80$ t in minutes, gives $T(t) = 20 + 60e^{-0{,}1t}$. With a step h, Euler gives $T_{n+1} = T_n - 0{,}1h(T_n - 20)$ in moments $t_n = nh$. For $h = 2$, on obtains 80,68,58,4. These are approximations of the continuous model at the moments 0.2,4; a law discrete physical given independently is not automatically Euler.

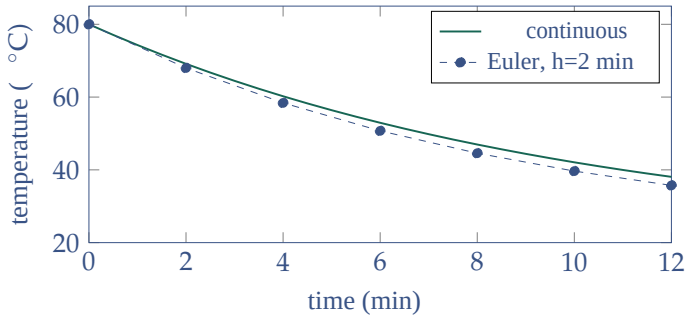


Figure 7.2. Read the deviation at t=6 min; segments connect successive approximations.

A6 · Choice of strategy Determine the first moment the continuous model reaches 50 degrees Celsius. Compare to the first value for Euler, with h=2 min, which is less than or equal to 50.
why the two results differ.

A first step: First solve an exponential equation; to Euler, write the formula of $T_n - 20$.

Wait a minute at the courthouse. Say what you mean by it, y' and t, then verify that the units of both members a differential equation is compatible.

Review postponed · S78. Complete the table of antiderivatives with trigonometric functions and control compositional factors.

Research activity

Cooling a drink · 40 min.

A drink initially at 80 °C is placed in a room held at 20 °C. For $t \geq 0$, note $T(t)$ its temperature in degrees Celsius, t the time elapsed in minutes. We admit the model $T'(t) = -0{,}1(T(t) - 20)$ and $T(0) = 80$.

1. Solve this initial problem and give expression to $T(t)$.
2. Study the direction of variation in T and determine its limit in $+\infty$. Interpret results.
3. Determine the first time the temperature is 40 °C. Give the exact value, then a value approached in minutes.
4. Apply two steps of the Euler method with a step of 1 minute. Compare the approximation obtained after two minutes at the exact value $T(2)$.
5. Indicate a concrete situation in which the hypothesis of a constant ambient temperature would no longer be suitable.